THE PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

ARUSHA REGION

FORM FOUR PRE-MOCK EXAMINATION

AGRICULTURE 01

CODE: 034

TIME: 3 HRS MARKING SCHEME YEAR: 2026

SECTION A

1. MULTIPLE CHOICE ITEMS (10 Marks)

| i | ii | iii | iv | v | vi | vii | viii | ix | x |
|---|---|---|---|---|---|---|---|---|---|
| A | D | C | A | C | C | A | B | A | C |

2. MATCHING ITEMS (6 Marks)

| i | ii | iii | iv | v | vi |
|---|---|---|---|---|---|
| D | F | A | G | H | C |

SECTION B (54 marks)

3. (a) (i) Sheep → RUMINANT ANIMALS

Hen → NON-RUMINANT ANIMAL (2 Marks)

(ii) Differences between Ruminant and Non ruminant animals (3 Marks)

| RUMINANT ANIMALS | NON-RUMINANT ANIMALS |
|---|---|
| → They have four compartment of the stomach (Polygastric) | → They have one simple stomach (Monogastric) |
| → Chew the cud (Regurgitate food) | → Do not chew the cud (NO regurgitation) |
| → There is no enzymatic digestion in the mouth | → There is enzymatic digestion start in the mouth |

| RUMINANT ANIMALS | NON-RUMINANT ANIMALS |
|---|---|
| → Have microorganism in the rumen that digests cellulose | → Digestion of cellulose takes place only in the caecum by microorganism |

3(a)(ii)

3 (b) Briefly explain the roles of the following parts of alimentary canal of goat

(i) Rumen → is the first stomach chamber that used as a temporary food storage and site where fermentation of ~~food~~ feedstuff takes place. (1 Marks)

(ii) Reticulum → is the second compartment chamber where filtration of solid and undigested food can be taken back to the rumen as well as sending nutrients and water to the omasum (1 Mark)

(iii) OMasum → is the third room called book with many folders where water and nutrients are squeezed to the abomasum (1 Mark)

(iv) Abomasum → is the fourth chamber called the true stomach where all nutrients and water is absorbed and chemically digestion (enzymes) is taking place (1 Mark)

4(a) The following that affects the productivity of soil

(i) Poor soil structure and texture
(ii) Presence of pests and diseases
(iii) Poor climatic condition (5 Marks)

4(a) (iv) Poor soil drainage and aeration
(v) The use of poor crops varieties
(vi) Poor microbial activities

4(b) Factors that determines the fertility of the soil
(i) Soil depth
(ii) Good microbial activities
(iii) Good soil structure and texture
(iv) Free from pests and diseases of soil
(v) Free from weeds of soil (4 Marks)
(vi) Good soil drainage and aeration
(vii) Correct pH
(viii) High level of soil nutrients.

5(a) Cultural Methods of pests and diseases control
(i) Crop rotation
(ii) Mixed cropping and intercropping
(iii) Use of clean planting materials
(iv) Timely planting
(v) Timely harvesting (6 Marks)
(vi) Proper spacing
(vii) Mulching
(ix) Use of resistant crop varieties

5(b) (i) Bricks → Used for making walls, columns, pavements and in some extent floors.

(ii) Aggregates → Used to make concrete for floors, walls, columns and beams for various farm structure.

(iii) Timber → Used for construction of doors, windows, roofs, Fencing ports, and portioning the livestock houses.

(iv) Thatch grass → Used for ~~thatching~~ thatching roofs, and also are used for construction of ~~specific~~ specialized production struction such as Mushroom production houses.

(2 Marks)

6(a) Five (5) sign of parturition in dairy cattle.

(i) Udder enlargement (Bagging up)
(ii) Relaxation of pelvic ligaments
(iii) Swelling of the ~~vulv~~ vulva
(iv) Discharge of mucus from the vulva
(v) Restlessness and isolation of cow
(vi) loss of appetites.

(5 Marks)

6(b) Sign of heat in farm animals

(i) Becomes restless
(ii) loss of appetites
(iii) Frequent urination
(iv) slight rise of body temperature
(v) swelling and reddish vulva
(vi) Tend to mount other animal

(5 Marks)

7(a) Factors to Consider when Constructing livestock houses

(i) Animal's natural behaviour
(ii) Animals environmental requirement
(iii) Topography and drainage
(iv) Soil type
(v) Accessibility
(vi) Protection from wind and Rain
(vii) Exposure to sun
(viii) Durability
(ix) Security of surrounding.

(6 Marks)

7(b) Livestock housing → Refers to Confinement of Animals in a purpose of build an enclosure

(1 Marks)

→ The farm animals should be provided ~~with~~ with shelter or house such as building protects the ~~livestock~~ livestock from rain, heat, predator as well as thieves

8(a)(i) Timely planting → Is the essential period for optimum crop yield, especially in annual crop. This is very important and it is facilitated by timely land preparation (1 Marks)

(ii) Why is emphasized

(i) Crops benefits from nitrogen flush which is available at the ~~beggining~~ beginning of the Rain

(ii) Crops get enough ~~moistr~~ moisture and (1 Marks)

$ (a) (ii) escape environmental stress
(iii) Crops escapes from serious pests attack
(iv) Crops escapes from ~~pes~~ diseases infestation
(v) Crops establish earlier than weeds hence smothering

$ (b) ~~Four~~ (4) ~~Five (5)~~ Factors to be considered in determining time for planting of particular crops
(i) Rainfall (The onset of rains)
(ii) weather condition at harvesting time
(iii) Prevalence of pests and diseases
(iv) Soil moisture contents
(v) Temperature
(vi) Market demand
(vii) Cropping system. (2 Marks)

Q1 (a)

| | AGRICULTURAL SECTOR | NON-AGRICULTURAL SECTOR |
|---|---|---|
| (i) | Climate dependent | → Are non-climate dependent |
| (ii) | Mainly concentrated on producing Raw materials | → Involved in manufacturing, Processing of raw materials |
| (iii) | Labour intense | → Do not depend much in labour either skilled |
| (iv) | Relatively low use of advanced technology | → It require high technology |
| (v) | Produces are perishable | → The item manufactured are durable and long lasting |
| (vi) | Difficult in product differentiation | → Easy in products differentiation |

(8 Marks)

9(b) Ways in which non-agricultural sector supports agricultural sector.

(i) Providing opportunities for employment and source of income for the labour force (2 Marks)

(ii) The growth of non-agricultural sector is an effective measure in addressing the emerging challenges of unemployment, poverty and out-migration of rural labour

## SECTION C (30 Marks)

10 Importances of horticulture

(i) Source of human nutrition. eg vitamins and minerals

(ii) They are good source of employment ~~and~~

(iii) They are good source of local and foreign income

(iv) They are source health benefits compound and medicines

(v) They are source of flavor and aroma in the diety

(vi) They have aesthetic value to the environment

(vii) They are source of Raw materials for industries.

1.1. Relationship between Agriculture with other Subjects

(i) Agriculture and biology

Biology studies Plants and Animals where by Agriculture uses this knowledges to grow crops and keeping animals

(ii) Agriculture and chemistry

chemistry help farmers to understand soil nutrient, fertilizers and pesticides

(iii) Agriculture and Mathematics

Mathematics is used to measure land, calculate profit, mix chemicals and preparing of animal feeds

(iv) Agriculture and geography

Geography helps farmers to understand climate, Rainfall, soil types and location

(v) Agriculture and language subjects

Language help farmers read instructions, write records and Communicate in markets

(vi) Agriculture and Computer.

Computer Science helps in record keeping, weather forecasting and modernizing the technology.

1.2 Way of increasing land value.

(i) Irrigation in a dry weather
(ii) Drainage of swamps
(iii) Afforestation
(iv) Addition of manure
(v) Use of fertilizers
(vi) Land reclamation